Vakint reduction benchmark

Cached RustRed versus the FORM / FMFT reduction route

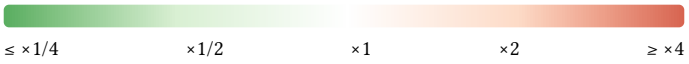
2026-10-24 · AMD EPYC 9754 128-Core Processor · CPU affinity 57

Reading guide. Bold values are RustRed times; FMFT times appear below. Multipliers are *RustRed ÷ FMFT*: green means faster, white parity, red slower. Both receive identical four-loop scalar inputs. The first physical denominator has power 2 (base) or 4 (held out); all others have power 1, auxiliaries 0.

Reduction time

Median seconds over 3 fresh processes per backend and case.

Topology	Base · power 2	Held out · power 4
H	**0.0359 s** FMFT 0.670 s ×0.054	**3.98 s** FMFT 7.37 s ×0.54
FG	**0.0450 s** FMFT 0.217 s ×0.21	**0.112 s** FMFT 0.430 s ×0.26
BMW	**0.0235 s** FMFT 0.181 s ×0.13	**0.997 s** FMFT 3.30 s ×0.30
X	**0.0822 s** FMFT 1.61 s ×0.051	**15.8 s** FMFT 21.0 s ×0.75



Method and correctness. Build profile: release (caller-declared); optimization/LTO settings are not inferred. rustc 1.98.1; FORM 5.0.0. Both retain masters in FMFT/MATAD normalization, with identical symbolic mass. Exact rational parity passed all eight cases at unit mass and renormalization scale, without truncating epsilon or using numerical tolerances.

Base cases hit prepared roots without applying rules. Held-out cases reuse cached descendants after root misses. Timing covers first public reduction calls, including FORM version checks, launch, I/O and parsing. Data loading, cache attachment, master evaluation and validation are excluded. Fresh processes run sequentially, alternating backends, with the recorded CPU affinity on a shared host. Filesystem pages are warm; this does not measure cold startup.

Selected validated run: `2026-10-24T06:59:51Z-8da99b9a5067`. Source: `8ef03d7fd755`. Data: `v0.1.2`. Samples and provenance: `data/results.json`. Refresh instructions: `../../AGENTS.md`.

Five-loop validation and benchmarks

Retained-master comparisons · functional evidence reported separately

Historical DOT ingress acceptance

2026-10-23 · one sequential debug-build pass. This checks graph definitions; it does not time IBP reduction, master evaluation, package loading or cache preparation.

Validation workload	Recorded result
Master graphs / dotted / with numerator	640 / 441 / 204
Exact routing, full numerators, native ingress and DOT round trips	Passed
Complete graph-validation pass	61.8 s
Per graph: minimum / median / maximum	0.0443 / 0.0955 / 0.175 s
Numerical master values / prepared 5L cache in this run	Absent / absent

Physical powers, signed indices, mass terms, full numerators, loop-basis slots (including zero) and integration Jacobians were checked exactly. These checks establish the stored graph definitions, without claiming that the catalogue is an independent master basis. One observed run supports no statistical speedup claim. Earlier interrupted checks are not a timing baseline.

Five-loop reduction observations

Retained-master workload	No persistent cache	With persistent cache
Complete quadratic UV fixture	Deferred	Deferred
Zero-momentum diagnostic	144005.8 s n = 1 · historical cohort Separate cohorts · no matched ratio	730.3 s n = 1 · final HEAVY · monitored Separate cohorts · no matched ratio
At least 12 held-out inputs	No matched suite timings	Functional controls reported separately

Reading guide. q0 cells are actual one-shot observations, with exact retained-result and guard validation. The user accepted the historical cold reference and waived new uncached repetitions. The cached observation uses the accepted final HEAVY file; its source cohort and monitoring differ from the historical producer. No matched three-sample median, cache-only ratio or performance colour is claimed. Final science, cache tiers and detailed timer boundaries appear on the delivery page. The quadratic example remains deferred.

Historical ingress: source `68f2263bf54d` (modified); RustRed `52c7463686b4` Symbolica `b6cca776f9dc`. Exact patch, executable, data and per-graph digests remain in JSON.

Historical compatibility record, separate from selected performance evidence: RustRed `91b72f8f5d42` Symbolica `9c8623ecb800`. Published upgrade: Symbolica `1205ec30890f`, FeynKit `d22e0f9fdbf8` acceptance was pending in that historical record. Full revisions: `data/results.json`.

Five-loop evidence and timing ledger

Selected report: `five-loop-2f0a5a637cf939141c8b1eeb`. Raw reports and their hashed launcher/compiler receipts remain in JSON. Qualified performance additionally binds complete result/guard files, unknown-master inventories and accepted window receipts. A successful retained expression alone does not establish catalogue closure or numerical evaluability.

Primary workloads

Complete quadratic UV fixture: Deferred; unmeasured in this scope.

Zero-momentum diagnostic: Complete exact final-file validation; n = 1 observations on the delivery page.

Historical finite-cache preparation

Producer `8ef03d7fd755` · original phase: `pilot`. Retained completed candidate; selection recorded separately as `828f61d0d6a6`. 2 / 2 selected roots complete; 25086 intermediate records; strict Reject policy. Compressed size: 2.342 MiB; decoded: 16593448 bytes (15.825 MiB). Original selected-pilot producer-driver elapsed: `1272.632297156` s, including loading and owner setup, not pure IBP or overall campaign time. Cache SHA-256: `cc5e6d80aacc737b81be1675a8aa02694b9ccaa1e4f31cc2d5a34199e4f1a19d`. Training roots, capture diagnostics and source/proof bindings remain in the unmodified build receipt.

Accepted native compatibility bridge `daa07861eaf7` · consumer cohort receipt `2b2a26eb74cc` · consumer workload `f022d2173668`. This does not establish a fresh target-suite pass, later-cohort compatibility or performance.

Separate rejected trial: 1 / 3 roots complete; `32613.349550541956` s whole attempt; no final gzip candidate.

This finite selection retains its original producer and acceptance. The cumulative cache from physical-example reductions is a separate record; see the physical capture supplement.

Timing boundaries. Stages overlap; never sum them. Owner preparation is included in saved-application/reduction times. `ibp_excluding_owner_seconds` means saved application minus owner preparation, including transport, combination and terminal substitution. Pure IBP, terminal-only, isolated gzip decode and primary owner-subtracted times remain unmeasured. Prepared-input E2E includes current input/data/cache loading and public reduction; generator preparation stays historical. Fresh/prepared scopes never mix. Initial loading uses Sparse; representation reload belongs to reduction. Missing events are "—", never zero. Full records remain in JSON.

Genuinely connected five-loop control

Supplement: same-engine warm reuse. This scalar ladder-pinch control is separate from the physical quadratic/q0 workloads and from held-out persistent-cache comparisons. Each fresh process makes one first public API call and then repeats it on the same engine, settings and input.

1 completed functional pair(s). These observations are non-isolated; no median, range or ratio is reported. Three qualified fresh pairs in one accepted window are required for timing aggregates.

First public call · observed seconds	Repeated call · observed seconds
`21.743564929`	`6.918791091`

Functional source `10b3a9a075fa` · executable `d2cebdd00c2a` · data `4c0877b63b2d`.

Scope. Default prepared 1L–4L caches are attached; no persistent 5L cache is attached before or after either call. On-demand owner preparation is included inside the API timer. Loading, input preparation, serialization, cache snapshots and parity checks are excluded. Reuse may involve several kinds of in-memory state; this is not a selected persistent-cache speedup. Pure IBP, owner-only and terminal-only durations remain unmeasured. These values never enter the cold-process or physical-workload medians. Family identity parity establishes neither catalogue closure nor numerical master availability.

Zero-momentum physical reduction

Completed cache capture · fresh reuse of the same input

Required physical scope: Zero-momentum diagnostic. **Deferred, unmeasured:** Complete quadratic UV fixture. User requested completion with only the zero-momentum physical example; the quadratic example remains deferred until a separate decision. Scope authorization: `b099866588db`. Historical records remain unchanged.

One completed fresh seen-input pair; functional evidence only. Complete retained output, canonical public nonzero-condition file, unknown-master inventory and family map agree. The fresh reuse wrote no new records and preserved the native snapshot identity.

Observed seconds	Capture producer	Fresh seen reuse
Complete public reduction	**144005.8 s**	**5749.0 s**
Supervised collector wall time	145299.8 s	5877.2 s

These are raw single observations, with possible concurrent historical workers. No median, speedup claim or performance colour is assigned. Producer reduction includes capture overhead; pure IBP time is unmeasured. Both processes load the same prepared scalar input; graph preparation is excluded.

Strict records captured	Seen Route hits / misses	Distinct masters / unknowns
2,229,270	**1,750 / 0**	**55 / 0**

Reuse applied 0 rules, with 0 Apply hits, 0 intermediate hits and 0 new records. Captured records are not a deduplicated-key census or the final HEAVY file size.

Producer source `d0e90fbf6537` actual Native archive `ba2a03c60969` ELF `490fd6492677` frozen source `02f29102ca27`. Optimization 3 and fat LTO are bound by the compiler receipt. This original producer cohort is separate from the current application/dependency build. Selected evidence: `physical-capture-717e66516cdcb6f5bd04a30b`. Complete source, executable, input, cache and result bindings remain in `data/results.json`.

Cache production policy. Preserve useful strict roots and completed descendants generated by the physical examples, together with every durable prefix and immutable base. The finite training driver alone does not define the final cumulative cache. Native packs have no 50 MiB publication target or 95 MiB aggregate cap; report actual storage and operational limits.

Seen-input reuse is separate from the independently frozen held-out suite and from same-engine RAM repeats. Protected held-out runs read an immutable snapshot; newly captured work goes to a separate delta. Completed descendants do not prove completion of an unfinished parent, stronger campaign coverage or numerical master availability.

Append accepted evidence with `merge_physical_capture_results.py`. Refresh and validation contract: `maintenance.typ`.

q0 coefficient assembly

Verified engineering observations · 28 October 2026

Coefficient-local retained-result assembly replaces the two whole-expression exact-combination passes in the earlier observation. All 58 master-product coefficients agree exactly with the authenticated original native baseline across 55 distinct masters; all 146678 canonical public guards and the full family map agree. Master values and numerical samples were never evaluated.

Observed seconds	Earlier · dc7364b	New · 2d70bfe Diagnostics OFF
Complete public reduction	6412.7 s	1048.3 s
Prepared input through result serialization	6542.1 s	1194.2 s
Supervised process wall time	6559.5 s	1205.7 s
Retained exact combination*	5815.2 s	231.7 s
Complete native saved call†	565.4 s	770.3 s

The matching prepared-input-to-serialization observations differ by ×5.48 (about 5.5×). **This is a single engineering comparison:** source/cache cohorts, instrumentation and concurrent background work differ. It supplies neither final three-sample medians nor a cache-only speedup claim. Both runs reuse the same prepared scalar input; original graph preparation is excluded.

*Earlier: outer 2941.7 s plus catalogue 2873.4 s. New: coefficient-local catalogue 231.7 s; the outer phase is absent, not a measured zero. †The saved call includes native cache lookup/decode, transport, weighted exact assembly and terminal substitution. Its observed duration increased. Pure IBP and terminal-only durations remain unmeasured.

Diagnostics ON confirms reuse. The separate fresh instrumented process took 1321.3 s through serialization, 1334.5 s supervised wall time, and 1102.8 s for the public call. It returned the identical 784045-byte result and complete guards as OFF, with 1750 Route hits, 0 misses, 0 new rule applications and 0 captured records. OFF/ON controls hit diagnostics; both attach a persistent cache.

New source `2d70bfe439d8` · Native `c2ca815cc906` · kernel `9cd57678d881` · rustc 1.98.1 (48a229cea 2026-09-01) · declared profile release. Original compiled cohorts and verifier-host limits remain in the raw receipts; these checks do not establish final Main-host acceptance.

Selected engineering record: `q0-engineering-5327526dec8cda942773dc57`. Hashed source index: `provenance/q0_engineering_evidence_index.json`. Raw benchmark, execution, exact comparison and all product proofs are appended in `data/results.json`. Earlier selections and all four-loop records remain unchanged.

Cache file engineering observations

HEAVY · one complete cold q0 process. Same executable, source `2d70bfe`, Native `c2ca815` and kernel `9cd5767` as the earlier INDEX run. All 58 coefficients, 55 masters, 146678 canonical public guards and the full result agree. Capture was enabled with delta reads disabled: 1750 Route hits, 0 misses, 0 new rules and 0 captured records.

Observed seconds	Earlier INDEX · OFF	HEAVY single FILE
Data / native cache loading	134.7 / 0.380	135.4 / 931.5
Complete public call	1048.3	777.0
Collector end to end	1194.2	2013.7
Inclusive controller elapsed	1205.7	2132.1
Native saved call / final coefficient combination	770.3 / 231.7	568.6 / 175.2

Public call 25.87% lower; cold collector end to end 68.63% higher. The lower native and CAS observations cannot isolate a cache-format benefit: concurrent work and capture policy differ. No causal format advantage, overall cold speedup, warm result or serialized three-run median is established.

Native FILE loading covers authentication, decode and indexing. The collector made two admitted SHA scans of the 96 GB file: its end-to-end timer includes the first scan and flush/snapshot, excludes close and final SHA; the controller includes both scans and cleanup. Their difference is not pure SHA time. SHA-only, decoder-only and pure IBP times are unmeasured. This legacy CLI used `file_identity=None` with Native framing/binding checks and before/after SHA, distinct from typed R3 acceptance.

LIGHT · one frozen tuning case, two utility pilots. A contains 54 Route records; B adds 2048 Apply records. Both fully match the original INDEX canonical result, guards and five families (68 masters, zero unknowns). Neither used held-out inputs.

Observed utility	A · 54 Route	B · 54 Route + 2048 Apply
Compressed bytes / native records	81907579 / 54	89635796 / 2102
Public / saved call (seconds)	470.7 / 453.6	510.1 / 493.7
Data / cache loading (seconds)	131.3 / 1.42	119.8 / 1.69
New rules / Route hits / Apply hits	52789 / 2 / 0	50381 / 2 / 2048
Closed capture records / directory bytes	117059 / 675020658	114225 / 666760498

B used fewer rules but took longer. The earlier full INDEX observation for this same case took 399.3 s and 4960 new rules (receipt `518d4963a340`). These fresh pilots ran alongside unrelated physical work; they do not establish a performance winner. The later user-directed LIGHT freeze is recorded on the delivery page. All new completed capture nodes remain preserved; neither run read its delta. Public timings include capture overhead.

Both native exports passed full reimport/count/ordered-guard checks within 95000000 compressed bytes. Their separate export helper measured encode+write 9.09 / 15.9 s, reimport 0.564 / 0.641 s and exact visitor comparison 2.28 / 8.02 s (A/B). Exporter identity receipts do not imply that the application supplied a typed Native file identity. Complete source cohorts, closed captures, raw phases and timer scopes remain in `data/results.json`. Source hashes in `provenance/cache_file_engineering_evidence_index.json`.

Five-loop delivery · zero momentum

Frozen LIGHT · cumulative HEAVY · final-file science complete

Scope and reference. The quadratic example remains deferred. The user accepted the historical uncached q0 reference, 144005.8 s (40.0 h, n = 1, original capture/cohort), and waived new uncached repetitions. Final cached observations retain their own cohort and n = 1 label: no matched median or cache-only ratio is claimed. The earlier 5.48× assembly observation concerns Vakint's coefficient-local source change, separately from cache format.

Cache tiers

Native file	Strict unique records	Compressed	Decoded
LIGHT	**2102**	**89.64 MB**	**104.66 MB**
HEAVY	**2257697**	**96.36 GB**	**116.19 GB**

LIGHT B: 54 Route + 2048 Apply records, accepted as sufficient by the user; selection/tuning ended. Complete Native reimport checked exact coefficients and original ordered cache guards. This does not identify a performance winner.

Published v0.1.2, then data-latest. Fresh immutable/latest downloads, fresh-process offline reuses and cache opt-out passed; all 640 uncomputed master getters passed in Community `c53b675` and Vakint `1f4f249`. Zero TCP connections applies to offline official-loader windows, excluding licensed startup.

Final-file cached q0 · one monitored process

Complete public reduction	Data / native FILE loading	Collector / controller E2E
730.3 s	117.4 / 851.8 s	1803.6 / 1906.7 s

Prepared scalar input; original graph preparation excluded. Native saved call 530.4 s; final coefficient combination 168.5 s, included in the public call. Loading includes authentication/decode/indexing; the collector and controller retain their different SHA/close boundaries. Pure IBP and isolated SHA/decode are unmeasured. Source `e4af69b` · Native `c2ca815` · kernel `9cd5767` · normal release profile. Actual Route hits 1750, misses 0, new rules 0. This seen-input check is separate from prior 96 GB engineering observations and held-out performance.

Exact science and monitor

Masters / product coefficients	Heartbeat freshness bound	Observed peak RAM
55 / 58	**2.491 s**	**9.23 GB**

All 146678 exported public guard Atoms agree in order, with mass, couplings, normalization and full family identities preserved. 1867 monitor samples cover the complete job, including loading, CAS, serialization and final SHA; native scientific-counter age remains separate. All workers exited.

RAM is the observed OS high-water mark, not an instantaneous allocation bound. Minimum sampled machine-available RAM 447.38 GB, above the 50 GB guard. Public guard order is separate from each Native cache record's original guard vector.

Final HEAVY complete: 3441543 original references across 13 closed sources; 2257697 unique records and 1183846 exact duplicates. Exhaustive reimport checked every coefficient and original ordered guard vector, including multiplicity. Native finalization census / export / reimport: 5584.4 / 13401.8 / 6101.0 s; these are cache-finalization phases, separate from original reduction/capture production.

HEAVY SHA-256 `79bb628313b63ef41428`…; LIGHT `708361954e61e853666d`… . Exact lengths, complete digests and local HEAVY path for independent hosting remain in the raw receipts. The old 96 GB snapshot is preserved separately; Native helper provenance is not a Vakint consumer build.

Held-out functional control: parent 30527, double dot/rank one; full value bytes, 142328 canonical public guards as a set, 68 used masters / 5 used families agree. Fresh replay: 1 requested-root hit, 0 new rules. Its 24988 strict records join HEAVY; LIGHT excludes them. No out-of-sample timing, unexported family-map equality or Native guard-vector comparison is inferred from this pair.

Hashed evidence: `q0_delivery_final_evidence_index.json`. All historical records and raw phase arrays remain in `data/results.json`. The 640 declared five-loop masters remain uncomputed; no master values were evaluated.

Ordered reduction · physical rung fixtures

Reduction math · exact engine parity · persistent caches disabled

Reading guide. All arms receive the same authenticated rung input and retain masters in MSbar. Net math includes **IBP/routing, request-local state and exact weighted coefficient assembly/grouping**. Data/program import, graph preparation, matching, initial lowering, normalization, numerics and serialization are excluded; measured lazy owner preparation is deducted. Qualified multipliers are *ordered ÷ DFS net math time*. The two DFS arms respectively retain and bypass per-run memoization. Ordered-cell color compares with the faster DFS arm; green means faster than both. Both ratios are printed. Without a bound fresh-process release protocol, observations are diagnostic and have no ratio. Gross times and preparation deductions remain in the raw record. Rows labeled n = 1 use a single actual DFS measurement; Ordered shows three-run median/range. Those single baselines have no measured run-to-run variability. Earlier balanced rows retain their original n = 3.

Case	DFS · memo	DFS · no memo	Ordered	Validation
1L · zero momentum	1.367 ms 1.365 ms–4.350 ms · n = 3	1.379 ms 1.330 ms–1.516 ms · n = 3	1.409 ms 1.385 ms–1.649 ms · n = 3 ÷ memo ×1.03 ÷ no memo ×1.02	Exact ✓ Numerical ✓ Fresh release
1L · quadratic	1.642 ms 1.554 ms–1.661 ms · n = 3	1.535 ms 1.517 ms–1.652 ms · n = 3	1.421 ms 1.413 ms–1.510 ms · n = 3 ÷ memo ×0.87 ÷ no memo ×0.93	Exact ✓ Numerical ✓ Fresh release
2L · zero momentum	6.252 ms 6.170 ms–6.416 ms · n = 3	7.094 ms 6.993 ms–7.160 ms · n = 3	5.653 ms 5.574 ms–5.810 ms · n = 3 ÷ memo ×0.90 ÷ no memo ×0.80	Exact ✓ Numerical ✓ Fresh release
2L · quadratic	61.932 ms 61.470 ms–67.326 ms · n = 3	100.2 ms 99.946 ms–103.5 ms · n = 3	9.810 ms 9.746 ms–10.263 ms · n = 3 ÷ memo ×0.16 ÷ no memo ×0.098	Exact ✓ Numerical ✓ Fresh release
3L · zero momentum	109.5 ms 108.9 ms–113.3 ms · n = 3	4.93 s 4.82 s–5.11 s · n = 3	68.567 ms 68.033 ms–71.418 ms · n = 3 ÷ memo ×0.63 ÷ no memo ×0.014	Exact ✓ Numerical ✓ Fresh release

Case	DFS · memo	DFS · no memo	Ordered	Validation
3L · quadratic	4.54 s n = 1 · single measurement	1601.6 s n = 1 · single measurement	274.4 ms 272.6 ms–277.0 ms · n = 3 ÷ memo ×0.060 ÷ no memo ×0.000171	Exact ✓ Numerical ✓ Fresh release

Public pipeline, collector and memory

Case	Public retained s	Collector s	Peak RSS	Phases
1L · zero momentum	0.0040 / 0.0040 / 0.0041	2.06 / 2.06 / 2.06	0.46 GiB / 0.46 GiB / 0.46 GiB	—
1L · quadratic	0.0050 / 0.0047 / 0.0046	2.07 / 2.07 / 2.08	0.46 GiB / 0.46 GiB / 0.46 GiB	—
2L · zero momentum	0.0125 / 0.0133 / 0.0179	2.09 / 2.08 / 2.13	0.46 GiB / 0.46 GiB / 0.46 GiB	—
2L · quadratic	0.0885 / 0.126 / 0.0972	2.29 / 2.36 / 2.32	0.46 GiB / 0.46 GiB / 0.46 GiB	—
3L · zero momentum	0.127 / 4.95 / 0.0893	1.74 / 11.5 / 1.75	0.24 GiB / 0.24 GiB / 0.24 GiB	—
3L · quadratic	4.72 / 1601.7 / 0.604	12.0 / 3204.5 / 3.16	0.24 GiB / 0.24 GiB / 0.24 GiB	—

Values above follow DFS memo / DFS no memo / Ordered order. Public retained time includes matching/lowering and finalization; it is not substituted for an unavailable math timer. Collector time begins after scalar-input preparation and includes numerical-reference generation when requested. RSS is the whole-process high-water mark, including earlier stages; a same-process pair shares that history. Phases refer to the last recorded native request, not all public-call channels. A dash means unavailable; isolated pure-IBP, routing/lowering and terminal-substitution timings remain null. No zero duration is imputed.

Prepared caches and previous-run results are not loaded. The actual intrinsic DFS memoization policy is recorded per arm. DFS no memo actually bypasses completed-result lookups, insertions and memo bookkeeping; its time is measured directly, never obtained by subtraction. Ordered's opaque factorization may use an isolated auxiliary memo; its primary frontier has no completed-result cache. Owner preparation uses one worker so its measured wall-time deduction is valid. Net math includes necessary terminal substitutions/projections and lower-factor reductions, not just the Native kernel.

Qualified samples disable monitoring in every arm; RAM guarding remains independent. Dashboard/snapshot runs retain their observer overhead and remain diagnostic, including the large five-loop examples.

Frontier, phase and cancellation counters are unmeasured in these monitoring-off timing samples and remain null in the raw records.

Correctness. Exact retained expressions agree. Numerical checks, where present, compare tracked component errors at unit UV mass, renormalization scale and coupling. The source-selected requested precision, achieved digits, warnings, data identity and payload hashes remain in each raw record. Error estimates are mathematical bounds only when their recorded quality is Bounded. Five-loop catalogue masters are uncomputed.

Source: `047e5ccd7309` Cargo lock: `9338911038ca`. Native / kernel: `b5e51865309d` / `9cd57678d881`. Data: v0.1.2. Full source-tree and manifest identities are preserved in the raw records.

These observations time the recorded source cohort. Later normalization/materialization cost fixes are not measured here; neither reduced math time nor the table's color implies a public-pipeline speedup.

Runs: `047e5ccd7-MSbar-1L-3L-balanced-20261030`, `047e5ccd7-MSbar-3L-quadratic-single-DFS-20261030`. Source/data/protocol and every raw sample: `data/results.json` → `ordered_rung_runs`. Refresh: `merge_ordered_rung_results.py` details in the maintenance contract and the ordered-engine guide. Historical FORM, DFS and cache cohorts above remain unchanged.